

EDUCATION

Nanyang Technological University (NTU), Singapore

Aug 2021 – Jun 2025

Bachelor of Computing (Hons) in Data Science and Artificial Intelligence (DSAI)

- Honours (Distinction) | CGPA: 4.20/5.00 | Minor Study: Business
- Relevant Modules: (1) Statistics (2) Data Structures & Algorithms (3) Algorithm Design & Analysis (4) Machine Learning (5) Neural Networks & Deep Learning (6) Natural Language Processing

ACHIEVEMENTS | AWARDS

- **Top 5% out of 496 participants, WorldQuant BRAIN Alphathon 2023** Sep 2023
- **Finalist, AWS Aifinity2023** Feb 2023 – Mar 2023
- **1st Runner Up Idea Award, Chat-bots for Mental Health 2022, Terra AI** Sep 2022

SKILLS

Languages: Proficient in English and Mandarin

Programming: Python, PyTorch, TensorFlow, scikit-learn, SQL, Linux, R, C, C++, Java, ReactJS, JavaScript, Git

Software Applications: AWS, Tableau, SQL Server, Firebase, Jupyter Notebook, Flutter, Figma, Microsoft Office

Certification: AWS Certified Cloud Practitioner, Google Advanced Data Analytics

WORK EXPERIENCE

Desay SV Singapore Pte Ltd.

Jan 2024 – Nov 2024

Algorithm Engineer Intern (Perception Team)

- Experimented and Fine-tuned road topology prediction model on large-scale dataset (OpenLane-V2, 2.1M+ instances), achieving 8.3% benchmark improvement over 3 months through ML techniques (Python, PyTorch)
- Built and integrated multitask perception system (topology prediction, semantic segmentation, lane and traffic elements detection), ensuring reproducibility and scalability by modular design with Git-based version control
- Designed APIs and generated intuitive visualization (bird's-eye overlays, video summaries) with open-source frameworks (OpenCV, MMDetection3D), improving model interpretability for data-driven decision-making

PROJECT

Natural Language Processing: Sentiment Classification of Movie Reviews

Oct 2024 – Nov 2024

Python | Pytorch | NLP | LLMs | Transfer Learning | Ensembling

- Implemented NLP models (BERT fine-tuning, LLaMA 2, bi-GRU, bi-LSTM) using Python and PyTorch, achieving 84.6% test accuracy in sentiment classification through transfer learning and ensemble methods
- Optimized text preprocessing pipelines with GloVe and FastText embeddings to improve tokenization efficiency, OOV handling, and semantic representation, enhancing model robustness for query and content understanding

Machine Learning: Prediction of Pawpularity Score for Pet Photos

Sep 2023 – Nov 2023

Python | PyTorch | scikit-learn | Deep Learning | Data Augmentation | Computer Vision | Ensembling

- Developed deep learning models combining pre-trained embeddings (CLIP, ResNeXt, ViT) with metadata to predict photo popularity, achieving top 7.5% RMSE (17.09) on Kaggle's "Pawpularity Contest" (9.9k+ images)
- Improved model performance and generalization through transfer learning, SVR-based ensembling, and data augmentation, leveraging PyTorch and scikit-learn for scalable experimentation and hyperparameter tuning

Database Design: E-Commerce Supplier Database Design and Optimization

Feb 2023 – Apr 2023

SQL | ETL | Relational Database | Query Optimization

- Designed and implemented normalized relational database schemas (3NF) with keys, constraints, and triggers to ensure data integrity and reliability, enabling accurate analytics on supplier transactions and customer records
- Optimized SQL queries in MS SQL Server to uncover seasonal sales trends and complaint resolution efficiency, delivering insights that could inform inventory planning, process improvements, and revenue growth strategies

LEADERSHIP EXPERIENCE | CO-CURRICULAR ACTIVITIES

CCDS Transition and Orientation Programme 2023, Honorary General Secretary

Sep 2022 – Aug 2023